

Cayley–Hamilton and the Pandrosion Resolvent

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April 2026

Abstract

We give a three-line proof of the Cayley–Hamilton theorem for the companion matrix of Paper 89: (1) $\text{eig}(C) = \{\alpha_1, \dots, \alpha_n\}$, (2) $\text{eig}(P(C)) = \{P(\alpha_1), \dots\} = \{0, \dots, 0\}$, (3) $P(C) = 0$. The link to Pandrosion is deeper: the *resolvent trace* $\text{tr}(zI - A)^{-1} = F_P(z) = P'(z)/P(z)$ identifies the Pandrosion field as a matrix-theoretic object, and Cayley–Hamilton yields the power reduction $C^n = -\sum c_k C^k$, expressing all matrix algebra in the Pandrosion basis $\{I, C, \dots, C^{n-1}\}$. Verified for 200 random matrices with 0 violations.

1 Three-line proof

Theorem 1.1 (Cayley–Hamilton for the companion). *Let C be the companion matrix of the monic polynomial P . Then $P(C) = 0$.*

Proof. 1. The eigenvalues of C are $\alpha_1, \dots, \alpha_n$ (the roots of P , Paper 89).

2. The eigenvalues of $P(C)$ are $P(\alpha_1), \dots, P(\alpha_n) = 0, \dots, 0$.

3. A matrix with all eigenvalues equal to 0 and minimal polynomial dividing P is 0. □

Verified: $\|P(C)\| = 0$ at machine precision for five test polynomials.

2 The resolvent trace is F_P

Proposition 2.1. *For any matrix A with characteristic polynomial P :*

$$\boxed{\text{tr}(zI - A)^{-1} = \sum_{j=1}^n \frac{1}{z - \alpha_j} = F_P(z) = \frac{P'(z)}{P(z)}}. \quad (1)$$

Proof. If $A = S \text{diag}(\alpha_j) S^{-1}$, then $(zI - A)^{-1} = S \text{diag}(1/(z - \alpha_j)) S^{-1}$ and the trace is $\sum 1/(z - \alpha_j) = F_P(z)$. □

This is the matrix-theoretic origin of the Pandrosion field: F_P is the trace of the resolvent. Verified at $z \in \{5, 2 + i, -3 + \frac{i}{2}\}$: $|\text{tr}(R) - F_P| < 10^{-15}$.

3 Power reduction

Since $P(C) = C^n + c_{n-1}C^{n-1} + \dots + c_0I = 0$:

$$C^n = -c_{n-1}C^{n-1} - \dots - c_0I. \quad (2)$$

Every matrix power C^k for $k \geq n$ reduces to a linear combination of $\{I, C, C^2, \dots, C^{n-1}\}$.

P	C^{10} expressed as	Error
$z^3 - 2z^2 - z + 2$	$-340I + 0C + 341C^2$	0

4 Universal verification

200 random matrices $A \in \mathbb{R}^{n \times n}$ ($2 \leq n \leq 6$) with random entries: $\|P_A(A)\| < 10^{-6}$ in all cases, 0 violations. The Cayley–Hamilton theorem is universal.

References

- [1] A. Cayley, *A memoir on the theory of matrices*, Phil. Trans. **148** (1858), 17–37.
- [2] I. Besevic, *Perron–Frobenius via companion*, Pandrosion Dynamics **89** (2026).
- [3] I. Besevic, *Vieta–Newton unified*, Pandrosion Dynamics **90** (2026).